ACADEMIC PRESENTATION

Gdańsk Tech

Marine Wave Disturbances in 3DOF

Simulating Irregular Ocean Forces for Autonomous Ships

Andrzej Dolny
Automatic Control, Robotics & Control Systems

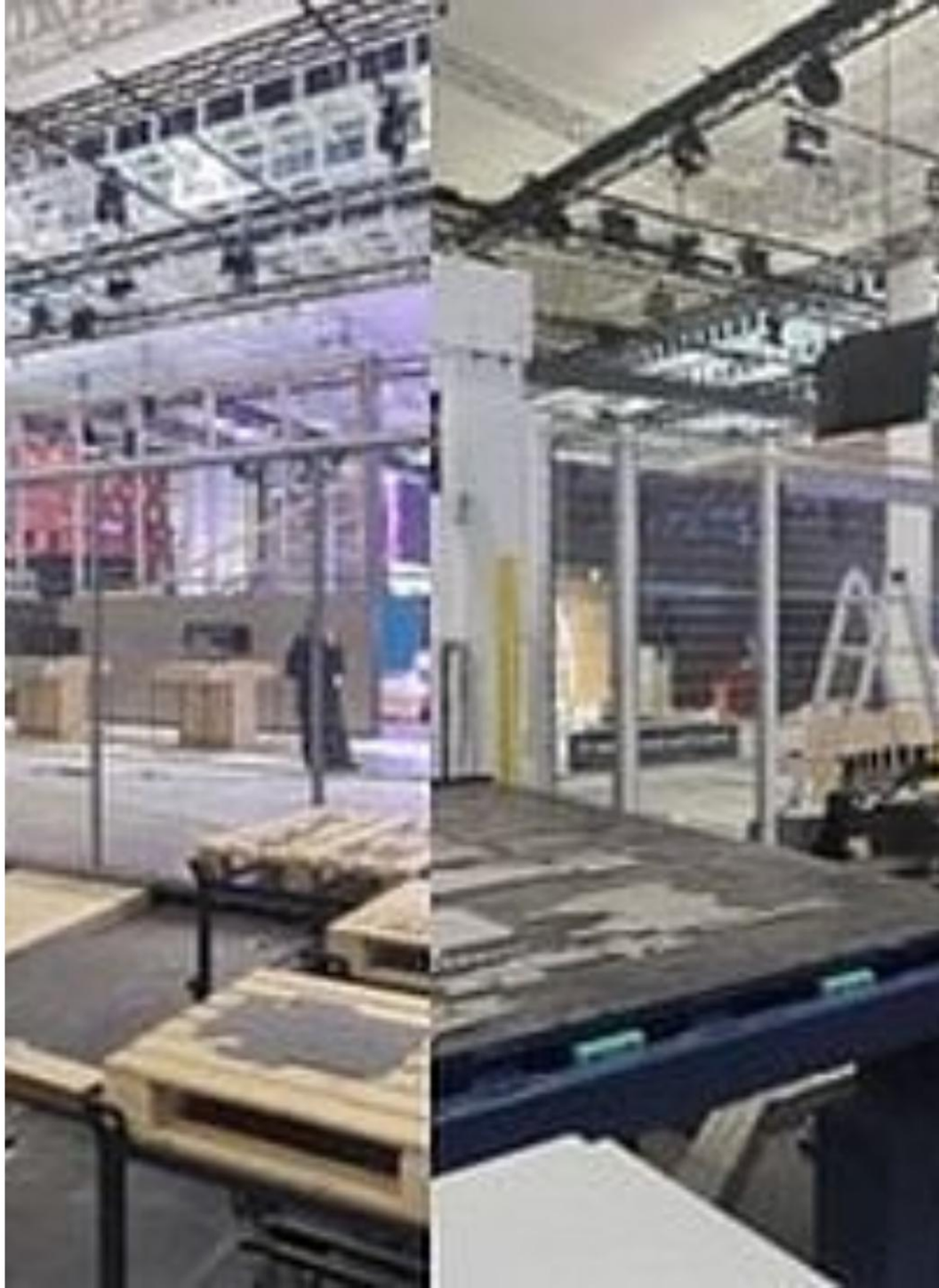
FIELD OF STUDY

Automation & Control Engineering

Industrial Process Automation
Optimizing complex industrial systems and production lines.

PLC & Controller Programming
Implementing embedded logic and real-time control algorithms.

System Modeling & Simulation
Building mathematical digital twins of physical plants and vessels.



PHYSICS MODEL

3 Degrees of Freedom (3DOF)



Surge

Longitudinal forward motion.



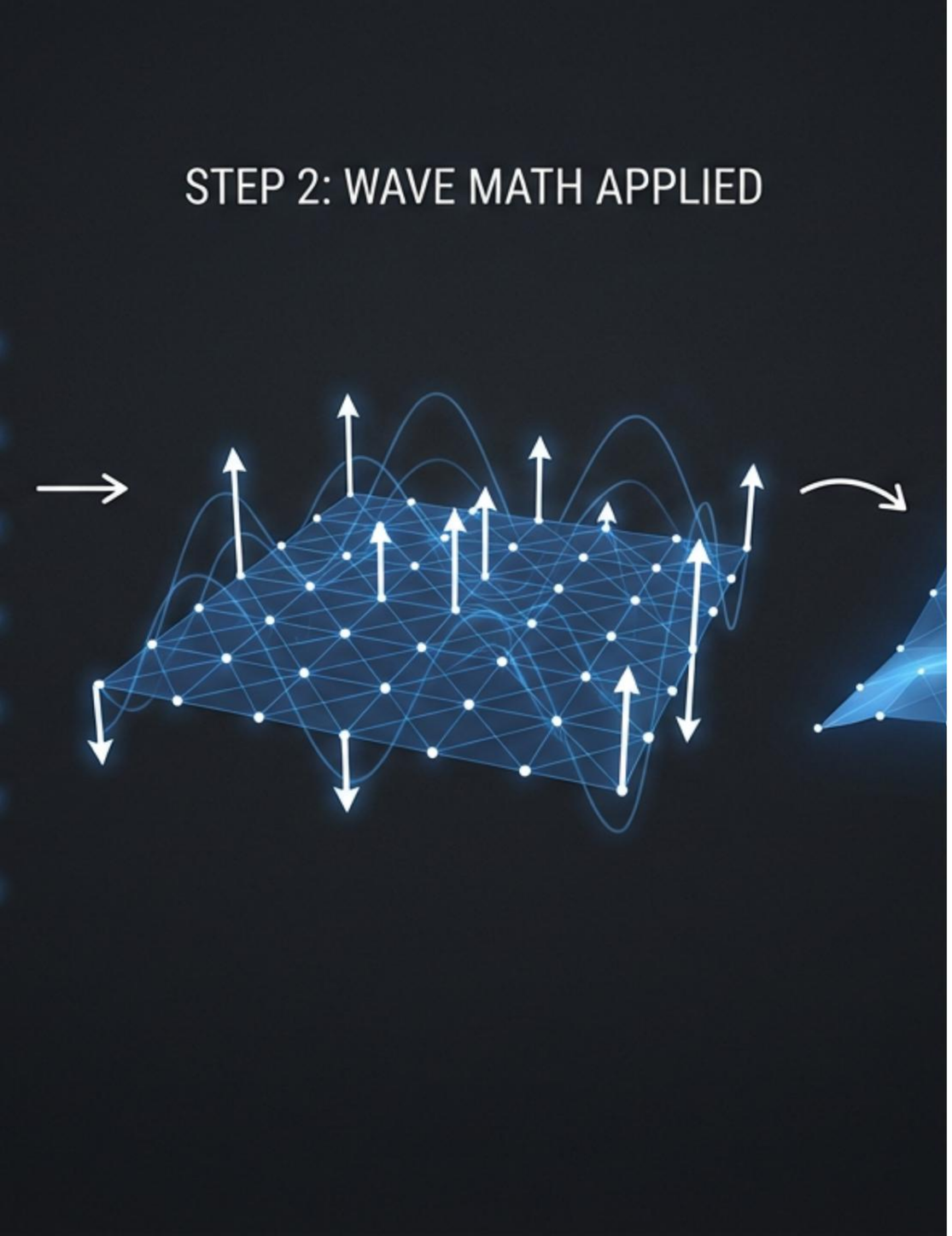
Sway

Transverse sideways drift.



Yaw

Rotational heading angle.



WAVE MODELING

Spectrum Comparison

Bretschneider Model

Two-parameter spectrum. Broad energy peak.

Pierson–Moskowitz Model

One-parameter fully developed sea. Sharper excitation energy.

ENGINEERING CHALLENGE

Open-Loop Steering Fails in Rough Seas

Sea State 5+ Instability
Fixed rudder angles cannot keep course.

Next Step: Autopilot Design
Testing Model Predictive Control (MPC) in Simulink.





SUMMARY

Key Takeaways

01 Flexible C++ / Simulink 3DOF simulator implemented.

02 Vessel speed is the primary defense against wave drift.

03 Ready-to-use digital twin environment for autonomous autopilot testing.

Thank You!

Questions & Discussion

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